

NikunjS91 / dark-phoenix-lunartech

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<> Code

🕒 Issues

🔗 Pull requests

👤 Agents

🔄 Actions

📁 Projects

📖 Wiki

🛡️ Security and quality

📊 Insights

⚙️ Settings

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Dark Phoenix assignment submission

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Code

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NikunjS91 and claude Add Playwright screenshots and embed them in README 5b6942d · last week

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📁 ai	DAY3: improve clip quality, rename clips, a...	2 weeks ago
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📄 README	Add Playwright screenshots and embed th...	last week
📄 WRITE_UP.md	DAY3: improve clip quality, rename clips, a...	2 weeks ago
📄 clips_manifest.json	DAY3: improve clip quality, rename clips, a...	2 weeks ago
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Dark Phoenix — AI Podcast Clipper

LunarTech.AI Assignment Submission Submitted by Nikunj Shetye

An end-to-end AI pipeline that transforms long-form podcast videos into short-form, watermarked, captioned clips — ready for social media distribution.

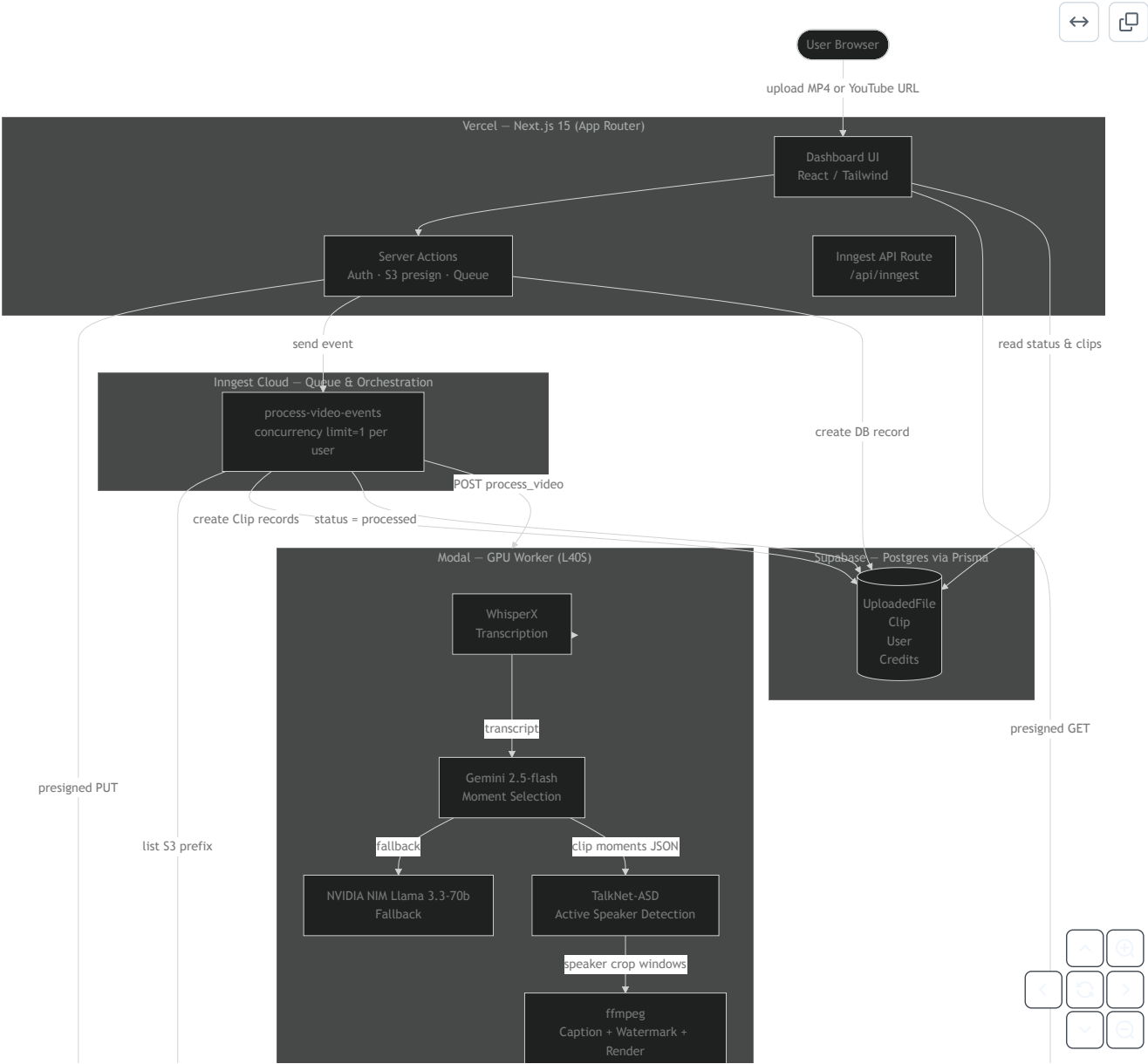
Live App: <https://dark-phoenix-gules.vercel.app> Reviewer Login: reviewer@dark-phoenix.com / Reviewer@2024

What It Does

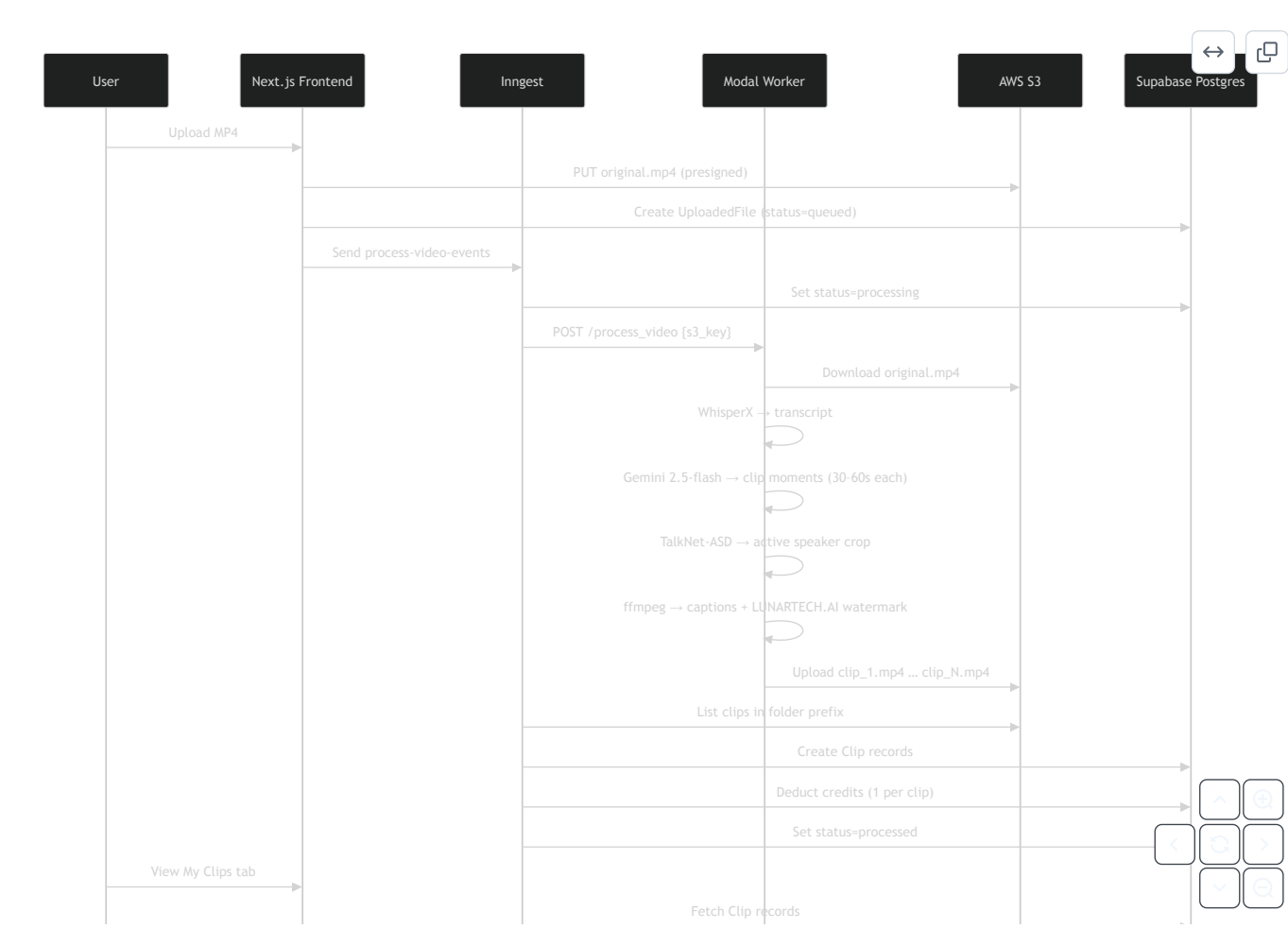
Upload a podcast video (or paste a YouTube URL). The system automatically:

- 1. Transcribes the audio using **WhisperX**
- 2. Selects the best moments using **Gemini 2.5-flash** (with NVIDIA NIM Llama 3.3-70b as fallback)
- 3. Detects the active speaker in each clip using **TalkNet-ASD**
- 4. Renders portrait-mode clips (9:16) with **burned-in captions** and a **LUNARTECH.AI watermark** via ffmpeg
- 5. Delivers the clips back to the dashboard for preview and download

System Architecture



Pipeline Detail



Key Implementation Decisions

Area	Choice	Reason
LLM for moment selection	Gemini 2.5-flash (primary) + NVIDIA NIM Llama 3.3-70b (fallback)	Gemini free-tier quota exhaustion during development; NIM as reliable fallback
Transcript sampling	Even-sample 2,500 of 9,856 segments (step=4)	Llama context limit is 131k tokens; full transcript exceeded it
Clip duration guard	<30s → extend to 45s; >60s → clamp to 60s	Ensures all clips are social-media viable length
Watermark	drawtext ffmpeg filter, top-right, 60% opacity, Anton font	Burned into video at export time — not a metadata overlay
Active speaker detection	TalkNet-ASD	Frame-level speaker activity detection for accurate crop windows
Queue	Inngest (concurrency limit=1 per userId)	Prevents GPU resource contention per user
Database	Supabase Postgres via Prisma + PgBouncer pooler	Serverless-safe connection pooling (port 6543, ?pgbouncer=true)

Repository Layout

└─ src/	
└─ app/	App Router pages
└─ actions/	Server actions (S3, generation, YouTube)
└─ components/	Dashboard UI, clip display
└─ inngest/	Queue function & client
└─ server/	Prisma DB client, Auth config
└─ prisma/schema.prisma	Data model
└─ ai-podcast-clipper-backend/	Python Modal app – GPU video processing
└─ main.py	Modal entrypoint (WhisperX, Gemini, ffmpeg)
└─ asd/	TalkNet active-speaker detection module

Running Locally

Prerequisites

- Node.js 20+, Python 3.11+
- [Modal](#) account (modal token new)
- AWS S3 bucket, Supabase project, Stripe account, Google AI Studio key

1. Environment

```
cp .env.example .env
# Fill in all values – see DEPLOYMENT.md for the full variable matrix
```



2. Frontend

```
cd ai-podcast-clipper-frontend
npm install
npx prisma db push
npm run dev          # http://localhost:3000
```



3. Backend (Modal)

```
cd ai-podcast-clipper-backend
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python setup_modal_secret.py # push .env into Modal Secret
modal deploy main.py        # prints endpoint URL → set as PROCESS_VIDEO_ENDPOINT
```



What Was Changed From the Original Repo

See [WRITE_UP.md](#) for the full technical write-up including every change made, failures encountered, and how each was fixed.

See [DEPLOYMENT.md](#) for the full deployment guide including all env vars, reviewer credentials, and infrastructure details.

Screenshots

Login

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Log in

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Dashboard — Upload & Queue

podcast/clipper

9983 creditsBuy morer

Podcast Clipper

Upload your podcast and get AI-generated clips instantly

Buy Credits

UploadMy Clips

Upload Podcast

Upload your audio or video file to generate clips

Drag and drop your file

or click to browse (MP4 up to 500MB)

Select File

Upload and Generate Clips

Or paste a YouTube URL

https://www.youtube.com/watch?v=...

Process Video

Queue status

Refresh

File	Uploaded	Status	Clips created	Actions
CS 691 Computer Science Capstone Project Sprint 1 (1).mp4	6/18/2026	Finished	5 clips	
sprint 2.mp4	6/18/2026	Finished	3 clips	
sprint 2.mp4	6/18/2026	Finished	3 clips	
YRvf00NooN8.mp4	6/15/2026	Finished	5 clips	
sprint 2.mp4	6/18/2026	Finished	No clips yet	
sprint 2.mp4	6/18/2026	Finished	No clips yet	
CS 691 Computer Science Capstone Project Sprint 1 (1).mp4	6/18/2026	Finished	No clips yet	

Releases

🏷 1 tags

[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 1

NikunjS91 Nikunj Shetye

Languages

● Python 59.4% ● TypeScript 35.1% ● CSS 3.1% ● JavaScript 2.4%

Suggested workflows

https://github.com/NikunjS91/dark-phoenix-lunartech

6/7

Based on your tech stack

 **Python Package using Anaconda** Configure

Create and test a Python package on multiple Python versions using Anaconda for package management.

 **Publish Python Package** Configure

Publish a Python Package to PyPI on release.

 **Django** Configure

Build and Test a Django Project

[More workflows](#)

[Dismiss suggestions](#)